

# Jianing Bai 白佳凝

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## EDUCATION

**Department of Computer Science and Engineering, University of Texas at Arlington, TX, USA** Aug 2026 – Present  
Ph.D. in Computer Science and Engineering

Core Courses: Design and Analysis of Algorithms / Advanced Multimedia, Graphics, and Image Processing

**College of Engineering, Peking University, Beijing, China** Sep 2020- Jul 2023

M.S. in Mechanical Engineering, Overall GPA: 3.07/4.0, Major GPA: 3.35/4.0

Core Courses: Artificial Intelligence: Methods and Applications (A) / Frontiers of Machine Learning (A)

**College of Information Science and Engineering, Hebei North University, Zhangjiakou, China** Sep 2015- Jul 2019

B.S. in Information Engineering, Overall GPA: 3.51/4.0, Major GPA: 3.86/4.0

Core Courses: Higher Mathematics (90) / Probability Theory and Mathematical Statistics (96) / Discrete Mathematics (93) / Computer Network (93) / Data Structure (91) / Network Communication Programming (93) / Network Database System (90)

## ACADEMIC FOCUS / RESEARCH INTEREST

Machine Learning, Medical Imaging, Multimodal Learning, Deep Reinforcement Learning, Control Optimization, Power Systems, Network Systems.

## PUBLICATIONS

1. **Jianing Bai**, Ren Wang, and Zuyi Li. [Physics-Constrained Backdoor Attacks on Power System Fault Localization](#). IEEE Power and Energy Society General Meeting 2023, IEEE.
2. **Jianing Bai**, Tianhao Zhang, Chen Wang, and Guangming Xie. [MA-CC: Cross-Layer Congestion Control via Multi-Agent Reinforcement Learning](#). Computing Conference 2023, Springer.
3. Yan Cao, Zhijun Jin, Rukai Zhu, Kouqi Liu, and **Jianing Bai**. [Comprehensive Evaluation of the Organic-Rich Saline Lacustrine Shale in the Lucaogou Formation, Jimusar Sag, Junggar Basin, NW China](#). Energy 2024, Elsevier.
4. Yan Cao\*, **Jianing Bai\***, Ruiyu Xu, Zhijun Jin, Jianguo Wu, Rukai Zhua, and Kouqi Liu. Predicting pore structure and oil-bearing characteristics in saline-alkaline lacustrine shale strata via transfer learning. (\* Co-first authors, under review).

## RESEARCH EXPERIENCES

**Deep Learning Based PET Image Analysis for Alzheimer's Disease Diagnosis** | University of Texas at Arlington | Research Assistant Sep 2026 – Present

Advisor: [Dajing Zhu](#), professor at Department of Computer Science and Engineering, University of Texas at Arlington

- Investigated deep learning approaches for PET image analysis to identify neuroimaging biomarkers associated with Alzheimer's disease and mild cognitive impairment.
- Developed machine learning and deep learning models for PET image processing, feature extraction, and disease classification using multimodal neuroimaging data.
- Explored advanced representation learning methods to improve the robustness and interpretability of PET-based diagnostic models under limited and heterogeneous imaging conditions.

**Multi-Agent Reinforcement Learning Based Network** | Peking University | Research Assistant Dec 2020 - May 2022

Advisor: [Guangming Xie](#), professor at College of Engineering, Peking University

- Researched and applied the multi-agent reinforcement learning algorithm to TCP and AQM to address the proposed cross-layer congestion control problem through cooperation.
- Designed a typical dumbbell-like network scenario under the ns3-gym simulator by modeling it as a multi-agent decision-making problem, which developed the application of MARL in the congestion control area.
- Proved that the proposed method outperformed typical rule-based and learning-based congestion control algorithms, potentially improving the adaptability and time variance of components of the Internet's communication infrastructure.

**Physics-Constrained Backdoor Attacks on Power System** | Illinois Tech | Research Assistant May 2022 - Jan 2023

Advisor: [Ren Wang](#), assistant professor at Department of Electrical and Computer Engineering, Illinois Institute of Technology

- Designed a novel physics-constrained backdoor attack strategy on DL-based fault line localization tasks in power systems.
- Considered different threat models where attackers could directly manipulate training data or only access measurements to improve the variety of the program; conducted fault localizations on the IEEE 68-bus power system..
- Demonstrated that the proposed physics-constrained backdoor attacks could fail the predictions using a small amount of

malicious data while maintaining a high accuracy on clean data.

**Transfer Learning for Shale Reservoir Property Prediction** | Peking University | Research Assistant Jan 2023 - Present

Advisor: [Jianguo Wu](#), professor at College of Engineering, Peking University

- Built and benchmarked a suite of regression models for predicting shale pore-structure and oil-bearing properties, improving generalization via GridSearchCV tuning and PCA-based feature reduction.
- Developed a transfer-learning workflow for regression with Mahalanobis-distance similarity, then training a transfer-enhanced Random Forest to boost performance under limited target labels.
- Demonstrated that transfer learning can markedly reduce target-domain data requirements in low-sample regimes.

## **AWARDS**

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1. First Prize, Triple-A Student Award (top 3%) 2016, 2017, 2018
2. Academic Excellence Award of Peking University (top 15%) 2023

## **WORKING EXPERIENCES**

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**Luchen Technology** | Beijing, China | Research Assistant | Jul 2023 – Jul 2026

- Implemented an Auxiliary Classifier GAN (ACGAN) in Python (PyTorch) for conditional image generation/recognition tasks.
- Packaged training/inference pipelines and experiment configurations to ensure reproducible runs and easier deployment.
- Improved training efficiency and scalability on multi-GPU/multi-node environments using data parallelism, pipeline parallelism, and tensor parallelism.
- Supported the team's distributed AI training/deployment workflow by aligning experiments with the company's platform/tooling (e.g., Colossal-AI / cloud infrastructure).

**Longmen Education Training Center** | Cangzhou, China | Middle School Mathematics Tutor | Jul 2019 - Aug 2020

- Delivered small-group and one-on-one tutoring for middle school students, focusing on algebra, geometry, and exam-oriented problem-solving.
- Designed targeted practice sets and mock exams aligned with school curricula and local entrance requirements
- Provided timely feedback on assignments and error-analysis notes, helping students build systematic approaches to common question types.

## **SKILLS**

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Tools: C/C++, Python (including PyTorch and TensorFlow), LaTeX, Linux OS, Matlab, Microsoft Office, Photoshop

Standardized Test Scores: TOEFL iBT: 104, GRE: 320